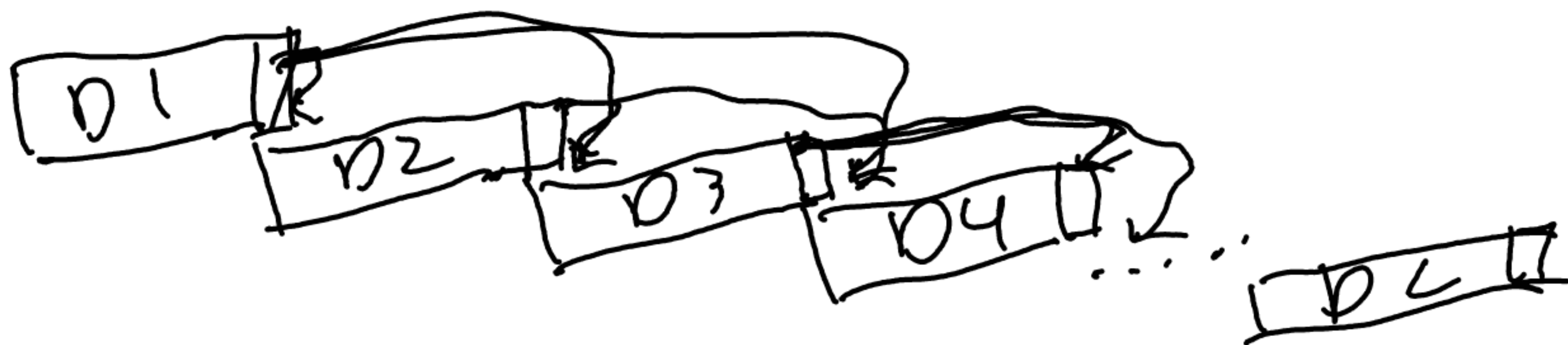


1) move static to dynamic - Arbitrary magic numbers

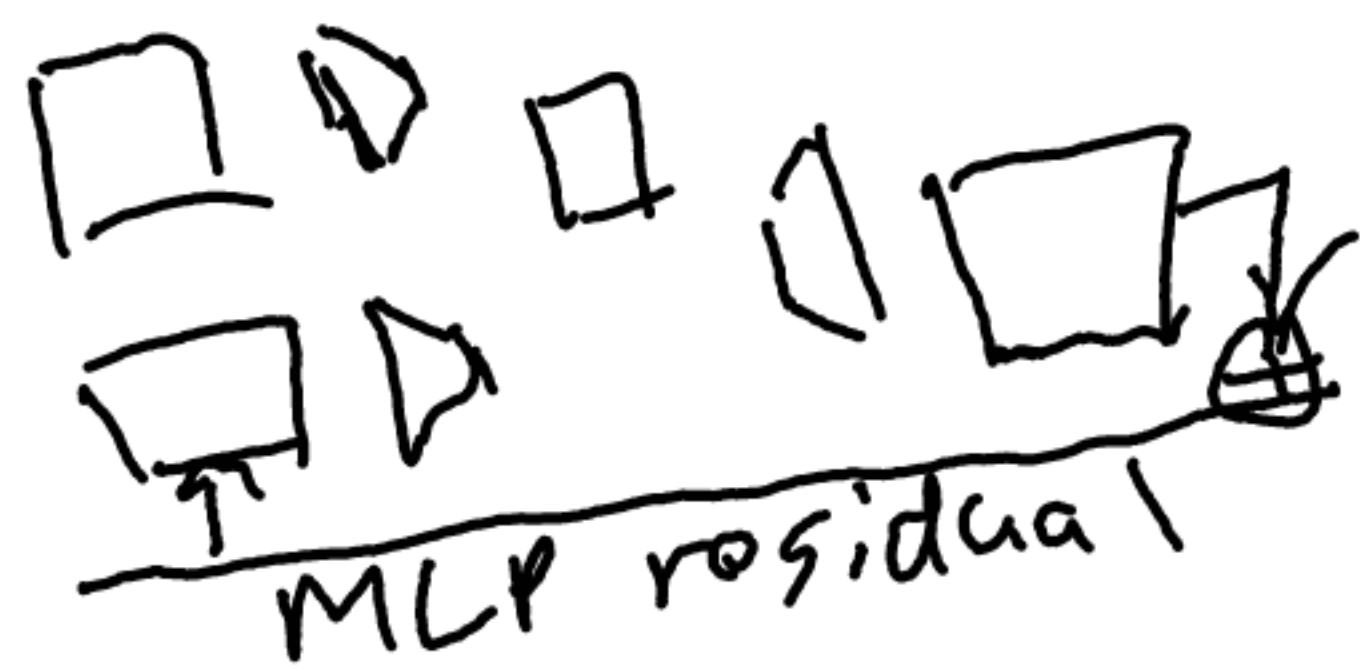
- top k $\rightarrow k$
- num layers / depth
- num experts
- num active experts
- dense vs sparse layers
- sparse vs mae
- transformer vs ssm vs conv vs mamba layers
- language size

2) compute reduction

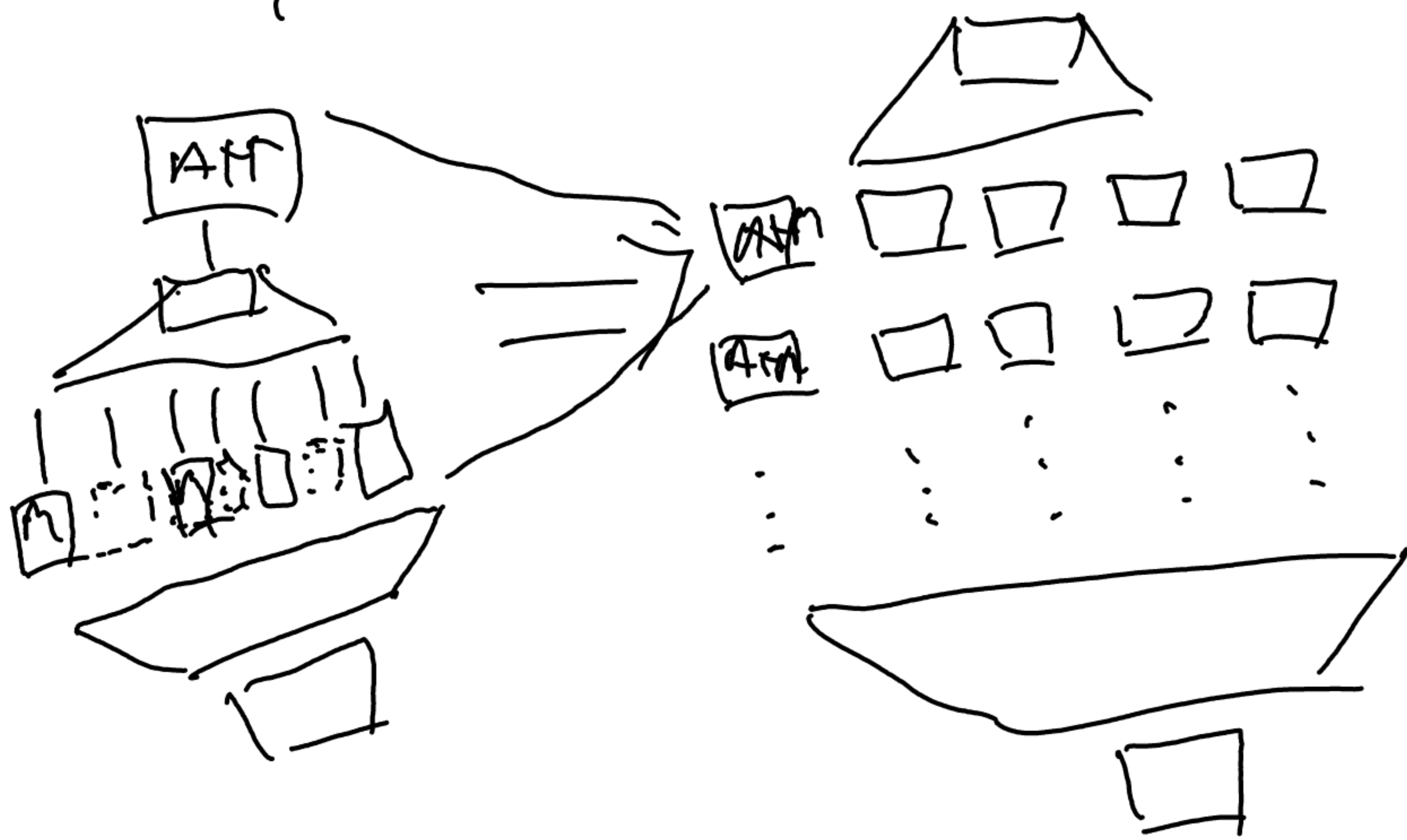
- hierarchical dictionary



- compressed dense layers



- new architectures
 - Layer MOE



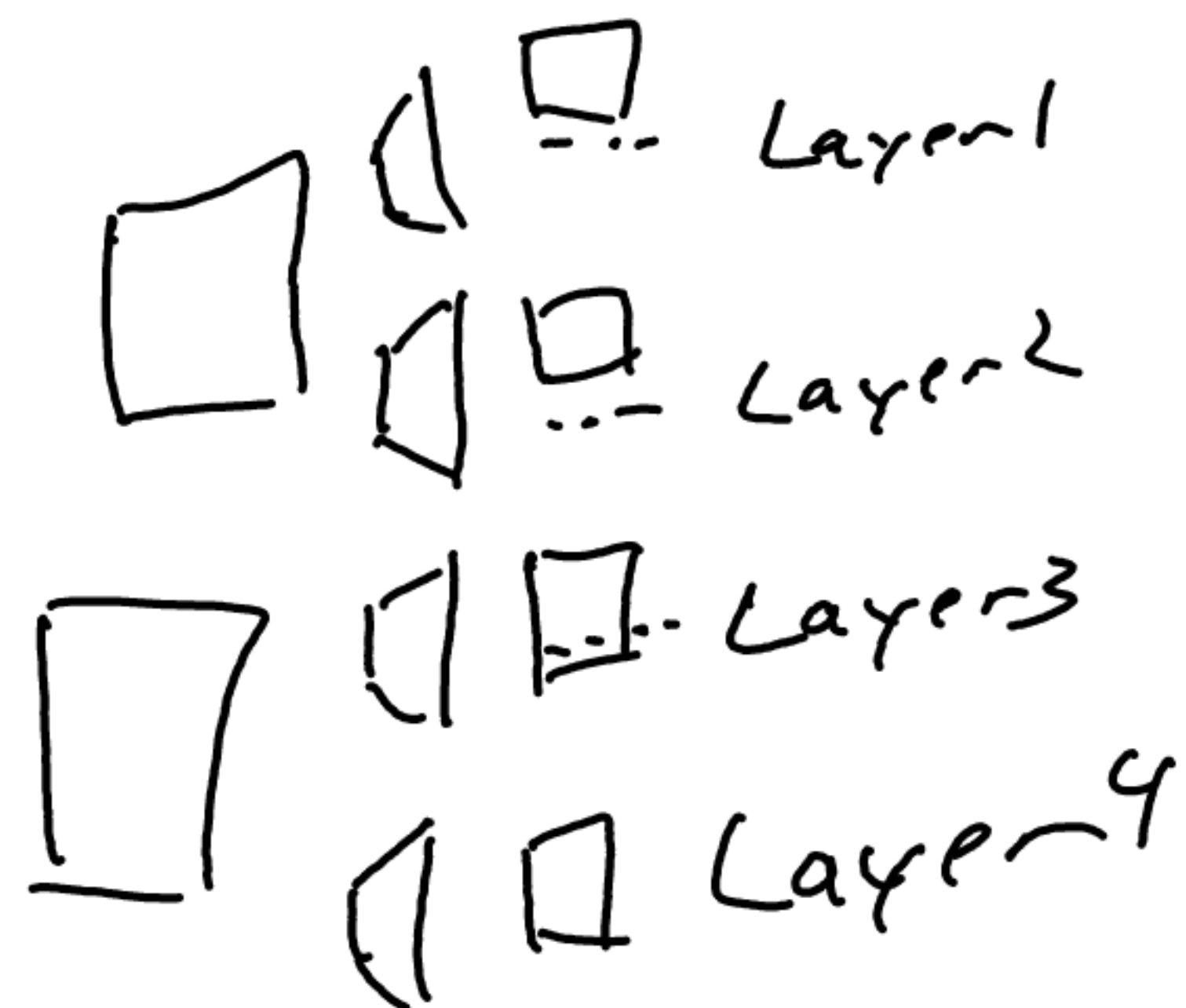
- swappable/replaceable experts
- dynamic expert router
- 1 - recursive internal state
- 2 - gated internal DAG

→ 1 + 2

- trainable activation function
- $$\text{Act} = \min(\max(f(x), \alpha), \beta)$$
- $$f(x) = ax + b$$
- or conv

- state machine core

x-layer
- shared weights



- lora everywhere

- compressed dictionary

- skip list layers

no residual

- flow control model

- multi-agent model
i.e.

mixture of models

Memory reduction

- TurboQuant at kv interface
- LSTM attention
 - ~~state-machine~~ grammar attention
- linked attention
- ragged window attention
- double attention
- block compressed weights
- block sparse weights
- dynamic type weights